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Influence of percent seed germination of three cabbage cultivars in different inoculum concentration of the bacterium *Xanthomonas campestris* pv. *campestris*

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Abstract : The effectiveness of inoculum concentration for seed germination in three different cabbage cultivars has been studied. Seeds were obtained from Kanke near Birsa Agriculture University, Ranchi. Black rot of crucifer caused by the bacterium *Xanthomonas campestris* pv. *campestris*(*Xcc*).This disease induces most effective symptoms in cabbage cultivars. The inoculum strength of 10^1, 10^2, 10^4 and 10^6 CFU/ml of *Xanthomonas campestris* pv. *campestris* were taken. Seeds were pin pricked and dipped in each of the four inoculum concentration separately for 5, 15 and 30 minutes for 1, 2, 3, 6, 9 and 12 days. The 5 min of inoculation dip was insufficient for affecting seed mortality. 15 and 30 min of inoculation dip had almost similar effect on seed mortality. The best inoculum concentration for affecting seed mortality was found 10^6 CFU/ml in the three cabbage cultivars Green Express, Krishna and Pusa drumhead respectively. Apparent seed mortality was observed for initial few days followed by overall delayed seed germination.

Key words :Inoculum, seed, germination, mortality, cabbage.

INTRODUCTION

The purpose of our investigation is in different inoculum concentration of the bacterium *Xanthomonas campestris* pv. *campestris,*effect of cabbage seed germination.Seeds of different cabbage varietes Green Express, Krishna and Pusa drumhead were taken from Kanke near Birsa Agriculture University,Ranchi. Because Black rot caused by the bacterium *Xanthomonas campestris* pv. *campestris* (*Xcc*), is considered the most prominent and most destructive disease of crucifers, infecting all cultivated varieties of brassicas near Birsa Agriculture University of Kanke region and can result in high losses of yield and quality. This disease induces V-shaped chlorotic to necrotic foliar lesions, vascular blackening, wilting, stunted growth, and stem rot symptoms on three cabbage cultivars i.e., Green Express, Krishna and Pusa drumhead.

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MATERIALS AND METHODS

In order to find out the effect of inoculum concentration and duration of inoculum dip on seed germination, one hundred (100) seeds each of cabbage cultivars Green Express, Krishna and Pusa drumhead were surface sterilized with 0.1% mercuric chloride. They were then pricked with sterilized needle. Four different concentrations of the bacterium having 10^1, 10^2, 10^4 and 10^6 CFU/ml was prepared separately. In 25 ml of each inoculum concentration 100 pin-pricked and previously sterilized seeds each of the three cabbage cultivars were kept. The duration of inoculum dip for seeds of each cultivar in all the four concentrations of inoculum was 5, 15 and 30 min. The percent seed germination for each inoculum concentration for every individual period of inoculum dip and for seeds of each of the cabbage cultivars

was calculated after 1,2, 3, 6, 9 & 12 days. The germination potency of the seed was checked with the help of blotter technique. Twenty five (25) seeds each under healthy and inoculated condition were kept in sterilized petridish on a blotter. In each petridish 10 ml of sterile distilled water was poured to keep it moist and at the end of each duration, the percent germination of seeds was estimated. The petridishes were kept in humid chamber for observation.

RESULTS AND DISCUSSION

An experiment was performed to find out the influence of inoculum concentration and period of inoculum dip on percent germination of cabbage seeds. The inoculum strength of 10^1, 10^2, 10^4 and 10^6 CFU/ml of *Xanthomonas campestris* pv. *campestris* were taken. One hundred seeds each of cabbage cultivars Green Express, Krishna and Pusa drumhead were taken. They were pin

Table-1: Percent seed germination of three cabbage cultivars under the influence of different inoculum concentration, varied periods of inoculum dip and during 1-12 days after inoculation of *xanthomonas campestris pv. campestris*

Inoculum Strength CFU/ml	Days After Inoculation	Percent Germination Of Seeds								
		Pusa drumhead			Krishna			Green Express		
		Period of Inoculum dip (Min)			Period of Inoculum dip (Min)			Period of Inoculum dip (Min)		
		5	15	30	5	15	30	5	15	30
10^6	1	4	-	-	4	-	-	8	-	-
	2	8	-	-	8	-	-	16	-	-
	3	12	4	4	16	4	-	20	4	-
	6	16	8	8	24	16	8	32	20	16
	9	20	16	12	36	20	24	36	28	24
	12	40	32	20	52	36	32	56	40	36
10^4	1	8	-	-	12	-	-	12	-	-
	2	16	-	-	16	-	-	20	-	-
	3	24	4	4	32	4	4	40	4	-
	6	32	16	12	36	24	20	52	20	16
	9	40	32	20	56	28	24	64	24	24
	12	44	36	28	76	36	28	80	32	32
10^2	1	12	-	-	12	-	-	16	-	-
	2	16	-	-	16	-	-	32	-	-
	3	28	4	-	32	4	-	60	4	-
	6	36	20	8	40	8	8	76	20	16
	9	44	28	16	56	28	28	80	32	40
	12	56	36	32	64	44	40	84	48	44
10^1	1	12	-	-	16	-	-	16	-	-
	2	20	4	-	32	-	-	36	4	-
	3	32	8	-	48	4	-	52	8	-
	6	56	20	8	56	28	20	76	24	12
	9	72	40	16	80	36	32	80	40	32
	12	76	44	36	84	52	36	88	52	40
Non Inocu-lated	1	12			12			12		
	2	36			40			36		
	3	40			44			52		
	6	64			64			76		
	9	88			80			92		
	12	100			100			100		

pricked and dipped in each of the four inoculum concentration separately for 5, 15 and 30 minutes. The non inoculated seeds were taken as control. The percent germination of seeds after 1, 2, 3, 6, 9 and 12 days was as certained. Results are presented in Table.

As would be evident 5 min. of inoculation dip was insufficient for affecting seed mortality. 15 and 30 min. of inoculation dip had almost similar effect on seed mortality. The best inoculum concentration for affecting seed mortality was 10^6 CFU/ml. The next concentration was 10^4 CFU/ml. It was interesting to note that up to 2 days after inoculation there was no seed germination. Delayed seed germination after 3 days, however took place. Thus high seed mortality was apparent up to 3 days after inoculation. Afterwards, between 6 to 12 days percent seed germination increased. It is apparent, therefore, that with declining bacterial numbers parallel to advance in days after inoculation, there is recovery of the seeds and percent germination increases.These trends were more or less similar for germinability of seeds of cabbage.

High percentage of seed infection in India was found[1]. Seed transmission of black rot of crucifers caused by *Xanthomonas campestris* pv. *campestris* too is a well known fact[2,3,4,5, 6,7,8,9&10]. Large number of *X. c. pv. vignicola* cells was recovered from the seed of *Vigna unguiculata*[11].

It has been reported for bacterial leaf blight disease of paddy that seeds are able to keep the causal bacteria, *Xanthomonas campestris* pv. *oryzae* alive for different duration and even up to the next sowing seasons[12,13,14&15].

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